# The MOND acceleration scale is one half of the dark-energy gravitational rate

Carl P. Zimmerman[1]

[1] *Briar Creek Tech, Charlotte, North Carolina, USA*

(Dated: July 31, 2026)

The acceleration scale $a_0$ that organizes galactic rotation curves is known to lie near $cH_0$. We record an exact closed form for it in terms of the dark-energy density alone, $a_0 = \frac{1}{2}c\sqrt{G\rho_\Lambda}$, equivalently $c^2\sqrt{\Lambda/32\pi}$. With the Planck 2018 $\rho_\Lambda$ this gives $9.43 \times 10^{-11}\,\mathrm{m\,s^{-2}}$, which is 0.70% from the value $9.36 \times 10^{-11}\,\mathrm{m\,s^{-2}}$ obtained by fitting 175 SPARC rotation curves at a stellar mass-to-light ratio $\Upsilon_{[3.6]} = 0.70$ (residual scatter 0.108 dex). The expression is parameter-free apart from the rational coefficient $\frac{1}{2}$. We state the limits of the claim explicitly. The interpolation used is identical to that of Milgrom [Phys. Lett. A **253**, 273 (1999)], which also *derives* $a_0 = 2cH_\Lambda$, a factor 11.58 larger; and the present coefficient is not observationally distinguishable from Milgrom's later empirical $cH_\Lambda/2\pi$, the two differing by 7.9% (0.018 dex in $g_{\rm obs}$, 16% of the fit scatter). The contribution is therefore one of *form*—an exact expression in $\rho_\Lambda$ with a rational coefficient—and not of improved fit. The coefficient is fitted, not derived. We give one falsifiable consequence: if $a_0 \propto \sqrt{\rho_\Lambda}$ then $a_0(z)$ is constant for $w = -1$ and rises to a maximum before declining for the DESI-preferred $w_0 > -1$, $w_a < 0$, in contrast to the monotonic rise predicted by $a_0 \propto cH(z)$.

## I. THE RELATION

Let $\rho_\Lambda \equiv \Lambda c^2/8\pi G$ be the dark-energy mass density and let $\sqrt{G\rho_\Lambda}$ denote the associated gravitational rate, the inverse of the dynamical time of a region of that density. The relation recorded here is

$$a_0 \;=\; \tfrac{1}{2}\,c\sqrt{G\rho_\Lambda} \;=\; c^2\sqrt{\frac{\Lambda}{32\pi}} \;=\; \frac{cH_\Lambda}{\sqrt{32\pi/3}}, \qquad (1)$$

where $H_\Lambda = c\sqrt{\Lambda/3} = \sqrt{8\pi G\rho_\Lambda/3}$. The three forms are algebraically identical. The reduction is immediate and worth displaying, because it shows that the factor $\sqrt{32\pi/3}$ carries no structural content:

$$\frac{cH_\Lambda}{\sqrt{32\pi/3}} = c\sqrt{\frac{8\pi G\rho_\Lambda}{3}}\sqrt{\frac{3}{32\pi}} = c\sqrt{\frac{G\rho_\Lambda}{4}} = \tfrac{1}{2}c\sqrt{G\rho_\Lambda}. \qquad (2)$$

Every factor of $\pi$ cancels, as does the 3. Accordingly $\sqrt{32\pi/3} = 5.78881$ should be read as the numerical conversion between $H_\Lambda$ and $\sqrt{G\rho_\Lambda}$ bookkeeping, and not as a geometric quantity to be explained. The entire content of Eq. (1) is the single rational coefficient $\frac{1}{2}$.

## II. RELATION TO PRIOR WORK

We place this before the numerical evaluation, since it bounds what can be claimed.

Milgrom introduced $a_0$ and noted its proximity to $cH_0$ [1]. The interpolation used throughout the present work,

$$g_{\rm obs}^2 = g_{\rm bar}^2 + a_0\, g_{\rm bar}, \qquad \nu(y) = \sqrt{1 + 1/y}, \qquad (3)$$

with $y = g_{\rm bar}/a_0$, is *identical* to Eq. (9) of Ref. [2]; it is not a variant of it. That same reference derives a value for the scale from a de Sitter–Unruh vacuum argument and obtains $a_0 = 2cH_\Lambda$, which exceeds Eq. (1) by the factor $2\sqrt{32\pi/3} = 11.58$. The de Sitter–Unruh route therefore does not leave the coefficient undetermined: it predicts one, and that prediction misses the measured scale by an order of magnitude. The present contribution is a renormalization of that coefficient to match data, not a new derivation. Comparable $2cH$-type coefficients are obtained in Refs. [3, 4].

The sharpest comparison is with the empirical $a_0 = cH_\Lambda/2\pi$ of Ref. [5]. Since $2\pi = 6.28319$ against $\sqrt{32\pi/3} = 5.78881$, the two coefficients differ by 7.9%, i.e. 0.036 dex in $a_0$. In the low-acceleration regime $g_{\rm obs} \simeq \sqrt{g_{\rm bar}a_0}$ this enters $g_{\rm obs}$ at half that, 0.018 dex, against a fit scatter of 0.108 dex (Sec. III). *The two coefficients are observationally indistinguishable at present precision.* We therefore make no claim of improved fit. What distinguishes Eq. (1) is that it is a closed form in $\rho_\Lambda$ with a rational coefficient, where $cH_\Lambda/2\pi$ is an empirical divisor.

Finally, we note for completeness that modified-inertia realizations of this phenomenology, their generic nonlocality, and the orbit-shape dependence of their effective interpolating function are established in Refs. [6, 7]; no result from that line is claimed here.

## III. NUMERICAL EVALUATION

Using Planck 2018 TT,TE,EE+lowE+lensing+BAO values $H_0 = 67.66\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_\Lambda = 0.6889$ [8] gives $\rho_\Lambda = 5.924\times10^{-27}\,\mathrm{kg\,m^{-3}}$, $H_\Lambda = 1.820\times10^{-18}\,\mathrm{s^{-1}}$, and

$$a_0 = \tfrac{1}{2}c\sqrt{G\rho_\Lambda} = 9.43 \times 10^{-11}\ \mathrm{m\,s^{-2}}. \qquad (4)$$

Fitting Eq. (3) to the 175 galaxies of the SPARC sample [9] with $a_0$ held at this value and a single global stellar mass-to-light ratio as the only free parameter returns $\Upsilon_{\rm disk} = 0.70$ (with $\Upsilon_{\rm bulge} = 1.4\,\Upsilon_{\rm disk} = 0.98$) at a residual radial-acceleration-relation scatter of 0.108 dex. For reference, the same fit with the conventional $a_0 = 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ and $\Upsilon_{\rm disk} = 0.5$ gives 0.122 dex. We stress that $a_0$ and $\Upsilon$ are degenerate in this fit, so the comparison should not be read as a measurement of $a_0$; the relevant statement is that $\Upsilon_{\rm disk} = 0.70$ lies inside the range 0.5–0.8 expected for Spitzer 3.6 $\mu$m stellar populations, so Eq. (1) does not require an implausible mass-to-light ratio.

One alternative normalization must be reported. Using the total density together with $cH_0$ in place of the pure-$\Lambda$ combination yields $a_0 = 1.13 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, 21% larger. The present fit does not discriminate between these; Sec. V gives an observable that does.

## IV. WHAT IS NOT CLAIMED

The coefficient $\frac{1}{2}$ is fitted. We attempted to fix it from ghost-freedom, unitarity, and a holographic degree-of-freedom count and did not succeed: the constructions that reproduce Eq. (3) leave the overall normalization free, and the one geometric candidate that appeared to fix it, $(3/8\pi)^{1/4} = 0.5878$, is the single-degree-of-freedom limit of a count with no independent microscopic justification. Equation (1) should therefore be read as a prediction of the *scaling* $a_0 \propto \sqrt{\rho_\Lambda}$ with an unexplained coefficient, and not as an explanation of the value of $a_0$. No claim is made here regarding particle physics or unification.

## V. A FALSIFIABLE CONSEQUENCE

The two normalizations of Sec. III differ in redshift dependence. If $a_0 \propto \sqrt{\rho_\Lambda}$ with a dark-energy equation of state $w(z) = w_0 + w_a z/(1+z)$, then

$$\frac{a_0(z)}{a_0(0)} = (1+z)^{\frac{3}{2}(1+w_0+w_a)} \exp\left[-\frac{3w_a z}{2(1+z)}\right]. \quad (5)$$

For $w_0 = -1$, $w_a = 0$ this is exactly constant. For the DESI-preferred region $w_0 > -1$, $w_a < 0$ it rises to a maximum and then declines; it is not a monotonic rise. By contrast $a_0 \propto cH(z)$ rises monotonically. Measurements of the acceleration scale at $z \gtrsim 1$ therefore discriminate between the normalizations, and a monotonic rise would exclude Eq. (1).

We note that at least one recent determination reports a rising $a_0(z)$, which is in tension with Eq. (5) on the pure-$\Lambda$ branch. That measurement is itself degenerate with $\Lambda$CDM assumptions in the mass modeling, and we record it as a live tension rather than as support.

## VI. KNOWN TENSIONS

We list these so that they are not overlooked.

*Clusters.* Equation (3) applied at cluster scales underpredicts the required acceleration. On uncorrected eRASS1 masses the median discrepancy factor at $R_{500}$ is 2.33, and the cluster-scale acceleration inferred from lensing, $2.0 \times 10^{-9}\,\mathrm{m\,s^{-2}}$, exceeds Eq. (1) by a factor 21.6. A weak-lensing mass recalibration reduces the former to roughly 1.6–1.8 but does not remove it. Because Eq. (1) gives a *lower* $a_0$ than the conventional value, this discrepancy is 13% worse here than for the conventional coefficient. The problem is shared with relativistic MOND theories generally and is unresolved.

*Solar system.* Applied exactly, Eq. (3) leaves a constant residual acceleration $a_0/2$ that exceeds Earth–Mars ranging bounds by three orders of magnitude. This is a property of the specific interpolation, not of Eq. (1): an interpolation with a faster Newtonian approach, $\mu(x) = x/\sqrt{1+x^2}$, leaves a residual $a_0^2/2g$ instead and reduces the anomaly by five orders of magnitude, at a cost of 0.003 dex in the SPARC fit. The word "exact" should not be attached to Eq. (3).

*Degeneracy.* As Sec. III notes, the rotation-curve data constrain the combination of $a_0$ and $\Upsilon$, not $a_0$ alone. Nothing here should be read as determining $a_0$ to better than the 7.9% separating Eq. (1) from $cH_\Lambda/2\pi$.

## VII. METHODS AND PROVENANCE

The relation Eq. (1) was arrived at independently by the author through three routes, which are recorded here for transparency about how the result was obtained.

1. An agentic automated-research harness [11], adapted by the author to search candidate dimensional combinations of $c$, $G$, $\Lambda$ and $H_0$ against the measured scale. (The cited harness is general-purpose infrastructure for running automated research agents; the search task applied here is the author's own.)

2. Symbolic regression over those dimensional combinations, in the spirit of the AI Feynman method of Udrescu and Tegmark [10], which recovered $a_0 \propto c\sqrt{G\rho_\Lambda}$ as the lowest-complexity expression consistent with the target value.

3. Direct fitting of Eq. (3) to the SPARC rotation-curve compilation [9], which fixes the coefficient at $\frac{1}{2}$ to the precision quoted in Sec. III.

Routes 1 and 2 identified the functional form; route 3 fixed the coefficient. We emphasize that none of the three constitutes a derivation, and that a coefficient obtained by fitting remains a fitted coefficient however it was found (Sec. IV). All numerical claims in this paper are reproduced by publicly available scripts that exit with a nonzero status if any internal consistency check fails [12].

## DATA AVAILABILITY

No new observational data were generated for this work. The rotation-curve analysis of Sec. III uses the publicly available SPARC compilation of Ref. [9], obtained from the archive maintained by its authors. The cosmological parameters are the published Planck 2018 values of Ref. [8]. All analysis code required to reproduce every numerical result quoted in this paper—the identity of Eq. (2), the fit of Sec. III, the evolution law of Eq. (5), and the cluster and solar-system factors of Sec. VII—is openly available in the repository of Ref. [12]. Each script performs internal consistency checks and exits with a nonzero status if any check fails.

## DISCLOSURE OF AI ASSISTANCE

Portions of the analysis, the numerical verification, and the drafting of this paper were carried out with the assistance of a large language model (Anthropic Claude). The author directed the work, specified and reviewed every load-bearing calculation, and takes full responsibility for the contents, including any errors. No artificial-intelligence system satisfies the criteria for authorship and none is listed as an author. Several intermediate claims produced during the work were subsequently found to be incorrect and were withdrawn prior to submission; the claims presented here are those that survived independent re-derivation.

## ACKNOWLEDGMENTS

The author thanks the SPARC team for making the rotation-curve compilation publicly available, and acknowledges the AI Feynman project for the symbolic-regression approach adopted in Sec. VII.

[1] M. Milgrom, Astrophys. J. **270**, 365 (1983).
[2] M. Milgrom, Phys. Lett. A **253**, 273 (1999).
[3] P. V. Pikhitsa, arXiv:1010.0318.
[4] F. R. Klinkhamer and M. Kopp, arXiv:1104.2022.
[5] M. Milgrom, Phys. Rev. D **102**, 084010 (2020).
[6] M. Milgrom, Ann. Phys. (N.Y.) **229**, 384 (1994).
[7] M. Milgrom, Phys. Rev. D **106**, 064060 (2022).
[8] Planck Collaboration, Astron. Astrophys. **641**, A6 (2020).
[9] F. Lelli, S. S. McGaugh, and J. M. Schombert, Astron. J. **152**, 157 (2016).
[10] S.-M. Udrescu and M. Tegmark, Sci. Adv. **6**, eaay2631 (2020).
[11] A. Karpathy, *autoresearch*, software repository (MIT license), `https://github.com/karpathy/autoresearch`.
[12] C. P. Zimmerman, analysis repository, `https://github.com/carlzimmerman/zimmerman-formula`.